Central Limit Theorem and normal distributions

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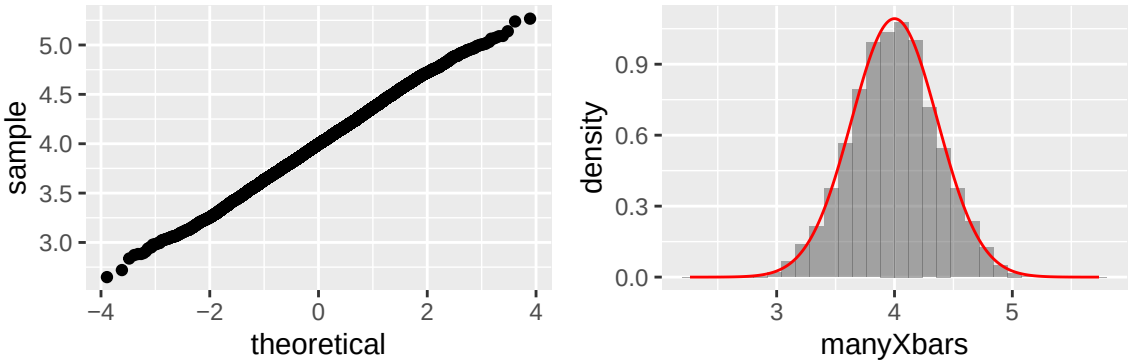
CLT and $\bar{X}$

We know $\bar{X}$ has a normal sampling distribution if we draw iid samples from a normal population. The CLT indicates this is true, when the sample size n is large enough, even when drawing from non-normal populations.

Drawing from a uniform distribution

Let us draw samples of size 10 from Unif$(2, 6)$.

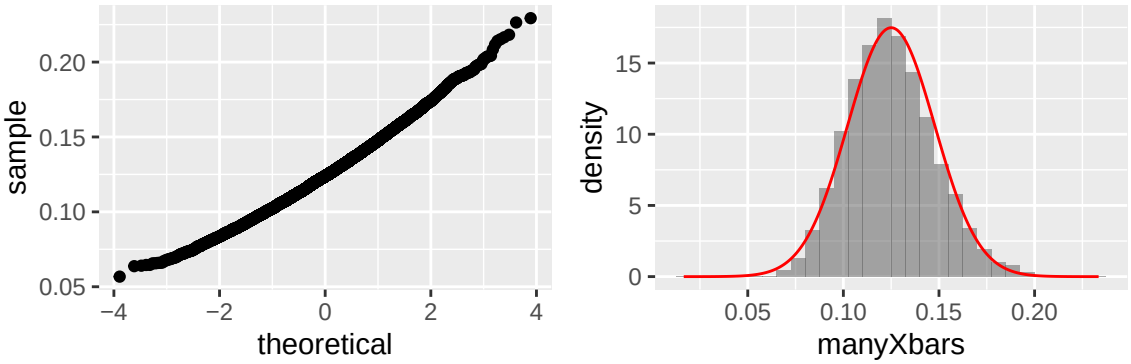
```
manyXbars = replicate(10000, mean(~ runif(10,2,6)))
p1 = gf_qq(~manyXbars)
p2 = gf_dhistogram(~manyXbars) |> gf_dist("norm", mean=4, sd=sqrt(4/30), color="red")
grid.arrange(p1, p2, ncol=2)
```



Drawing from an exponential distribution

Let us draw samples of size 10 from Exp(8).

```
n=30
manyXbars = replicate(10000, mean(~ rexp(n,8)))
p1 = gf_qq(~manyXbars)
p2 = gf_dhistogram(~manyXbars) |> gf_dist("norm", mean=1/8, sd=1/(8*sqrt(n)), color="red")
grid.arrange(p1, p2, ncol=2)
```



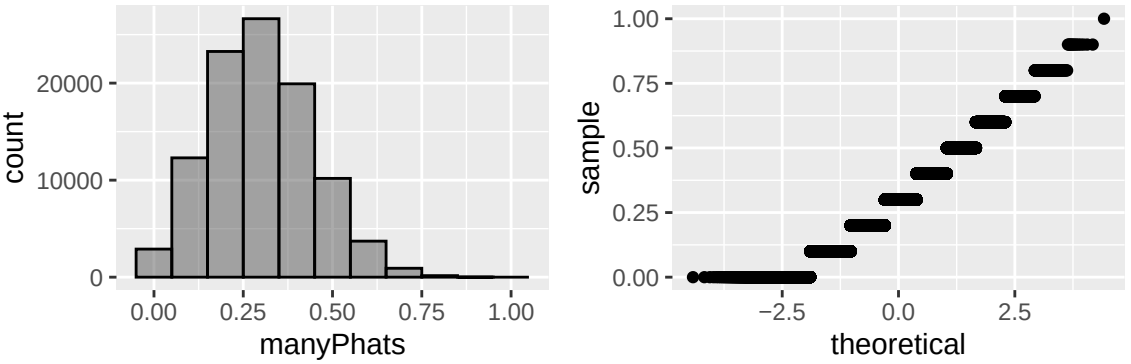
A long-accepted rule has been that $n \geq 30$ is sufficiently large to conclude $\bar{X}$ has a normal distribution, though examples like this one call into question whether this is true when drawing from a highly skewed population.

CLT and $\hat{p}$

iid draws from Binom$(1, \pi)$

First, I drawn $n = 10$ times and calculate the mean, which corresponds to the sample proportion $\hat{p}$:

```
n=10
manyPhats = replicate(100000, mean(~rbinom(n, 1, 0.3)))
p1 = gf_histogram(~manyPhats, binwidth=0.1, color="black")
p2 = gf_qq(~manyPhats)
grid.arrange(p1,p2,ncol=2)
```



The sampling distribution of $\hat{p}$ does not appear to be especially normal. Note that I used the value $\pi = 0.3$, and so

$$n\pi = 3 \qquad \text{and} \qquad n(1-\pi) = 7.$$

If I replace the `mean()` command by `sum()` in the code above, I'll be producing a simulated sampling distribution of X, the *count* (instead of the *proportion*) of successes. The last run illustrates the sampling distribution does *not* need to be normal. Nevertheless, this rule regarding the interplay between sample size and π, namely

$$n\pi \geq 10 \qquad \text{and} \qquad n(1-\pi) \geq 10,$$

is widely accepted as sufficient to consider the sampling distributions approximately normal for both $\hat{p}$ and $X \sim \text{Binom}(n, \pi)$:

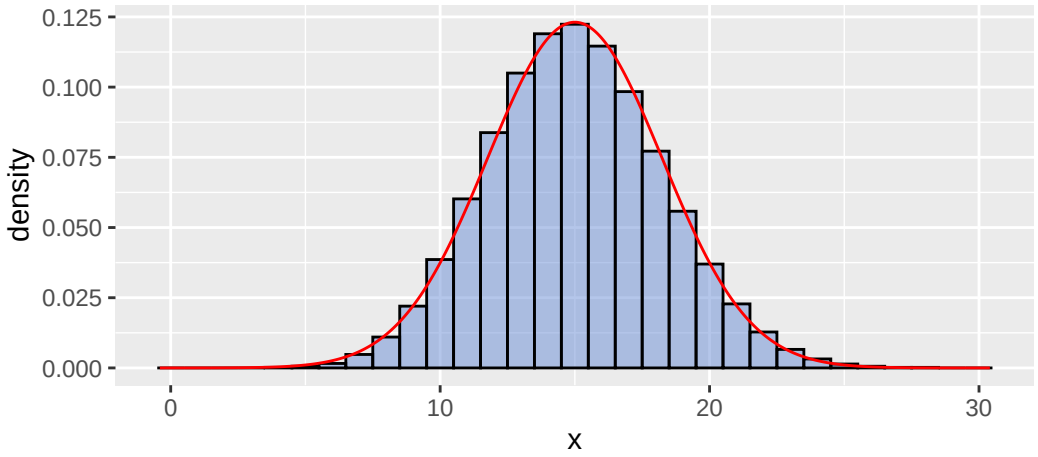
- $\hat{p} = \frac{1}{n}(X_1 + \cdots + X_n)$ has an approximate normal distribution $\text{Norm}(\pi, \sqrt{\frac{\pi(1-pi)}{n}})$, and
- $X = n\hat{p}$ is approximately distributed as $\text{Norm}(n\pi, \sqrt{n\pi(1-pi)})$,

when $n\pi \geq 10$ and $n(1-\pi) \geq 10$, assuming that the $X_i \sim \text{Binom}(1, \pi)$ are independent.

Continuity correction

The rule, $n\pi \geq 10$ and $n(1-\pi) \geq 10$ is met when $n = 50$ and $\pi = 0.3$. This should, by the statements above, make it viable to obtain probabilities for $X \sim \text{Binom}(50, 0.3)$ from `pnorm()` where the normal distribution $\text{Norm}(50, 0.3)$ is assumed. See the figure, which overlays the two distributions.

```
gf_dist("binom", size=50, prob=0.3, kind="histogram", binwidth=1, color="black",
        fill="#6B8ED6") |>
  gf_dist("norm", mean=15, sd=sqrt(15*.7), color="red")
```



Note that the binomial cdf function for values up to and including 10 disagrees with the normal cdf function.

```
pbinom(10,50,.3)
```

```
[1] 0.07885062
```

```
pnorm(10, 15, sqrt(50*.3*.7))
```

```
[1] 0.06141132
```

The picture, itself, suggests that replacing the discrete probabilities (blue rectangles) with continuous ones (area under the red curve) will lead to errors, much as calculus students learn to have discrepancies between true areas under curves and the approximations obtained using Riemann sums. That we might undertake some sort of *continuity correction* seems natural.

Just imagine, for the moment, computing the probability of the value $X = 10$. An error-free way, using the binomial cdf, is

```
pbinom(10,50,0.3) - pbinom(9,50,0.3)
```

```
[1] 0.03861899
```

Attempting to approximate this value using the normal cdf, you can see that

```
pnorm(10, 15, sqrt(50*0.3*0.7))- pnorm(10, 15, sqrt(50*0.3*0.7))
```

is a feeble effort giving zero, and

```
pnorm(10, 15, sqrt(50*0.3*0.7))- pnorm(9, 15, sqrt(50*0.3*0.7))
```

though an improvement, seems to be the wrong width-one section of area under the red curve. It would be better to get the area between 9.5 and 10.5. Similarly, `pbinom(10, 50, 0.3)` should be better-approximated using the normal probability up to 10.5 instead of 10:

```
pnorm(10.5, 15, sqrt(50*0.3*0.7))
```

```
[1] 0.08245741
```

Indeed, 0.082 is closer to 0.079 than 0.061.